

Results and Discussion

Segmentation Model

Training Loss: The given data presents the training loss of the segmentation model per epoch.

- The training loss measure's a model's performance during training
- Loss Function
- Improving Model
- Higher to Lower Loss
- Plateau near 0.001
- No overfitting

Validation Loss: Evaluates segmentation model's performance on the validation set during training

- Similar trend to training loss
- Downward Trend
- Minimum of 0.0036
- Good generalizability
- Slight overfitting

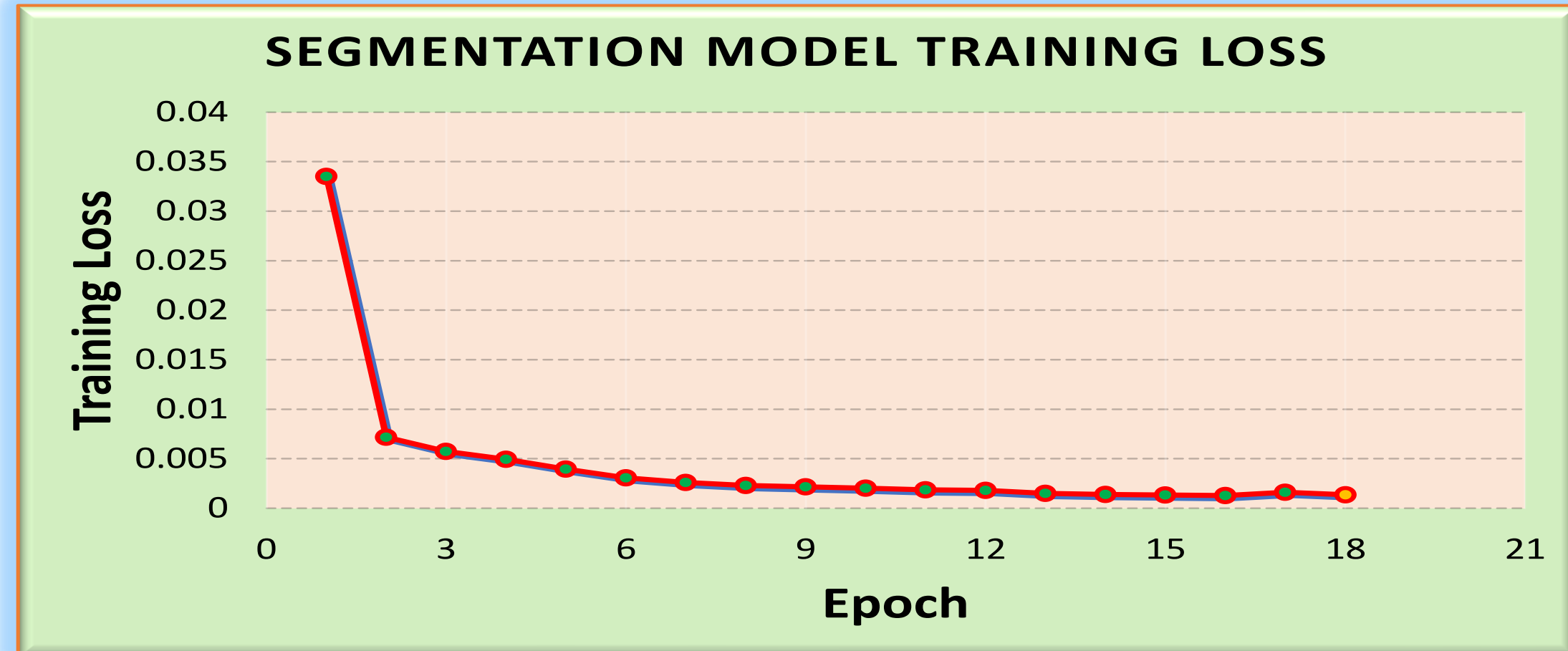


Figure 12 . Segmentation Model Results - Training Loss

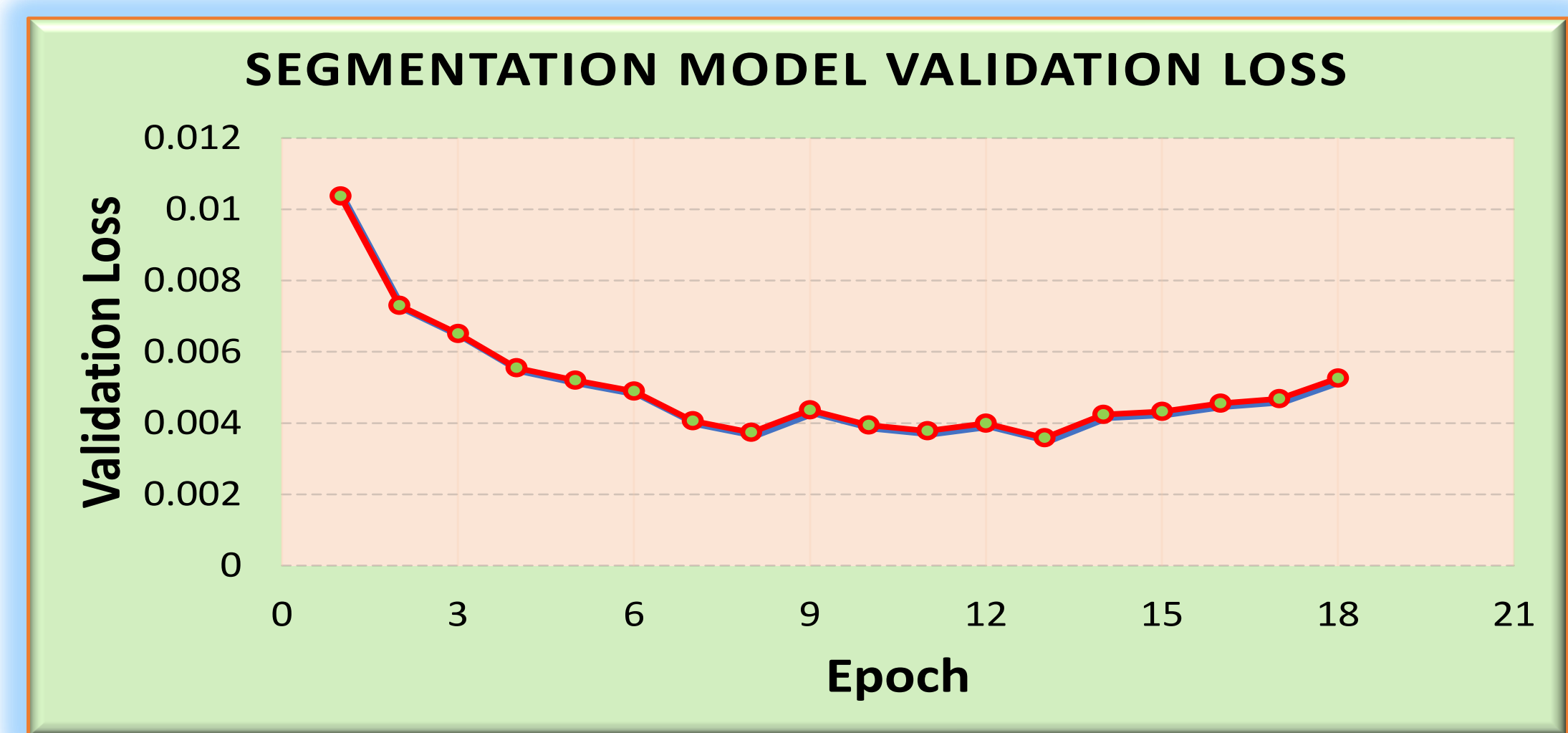


Figure 13 . Segmentation Model Results - Validation Loss.

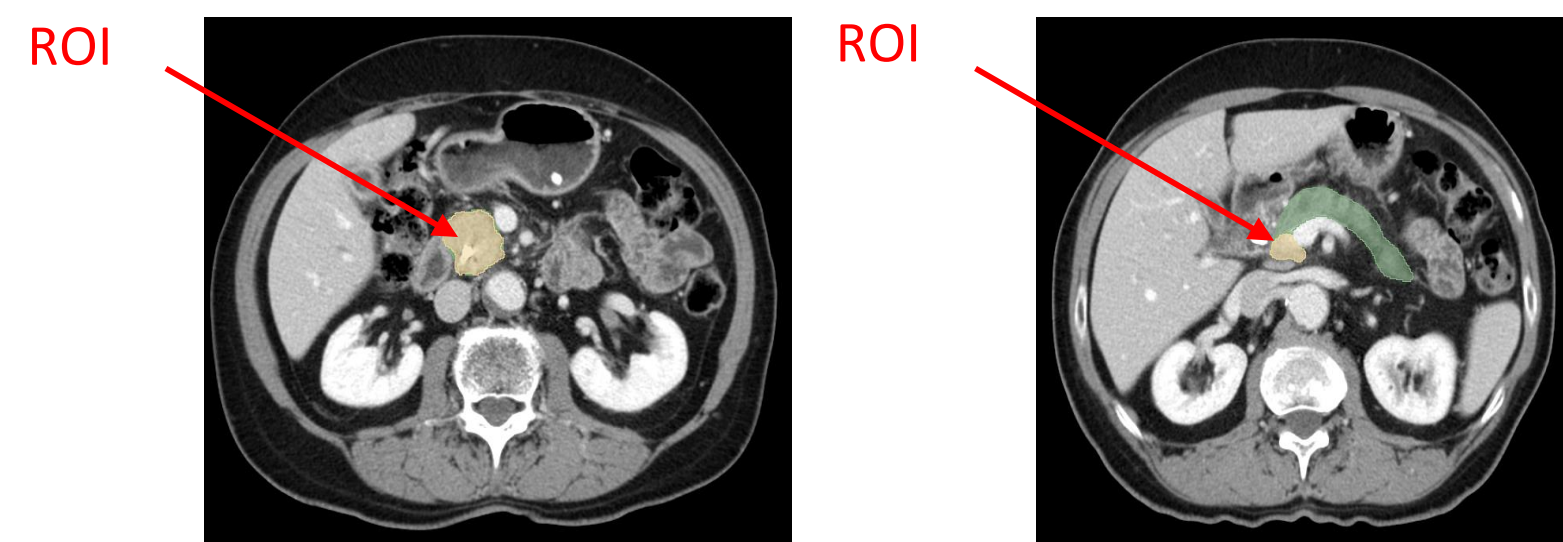


Figure 14. Image Segmentation Model sample outcome after segmenting the Region of Interest (ROI).

True Negative (TN)	False Negative (FN)
87.5%	12.5%
False Positive (FP)	True Positive
12.5%	87.5%

PREDICTION	ACTUAL	
	Negative	Positive
	TRUE NEGATIVE	FALSE NEGATIVE
Positive	FALSE POSITIVE	TRUE POSITIVE

Confusion Matrix

Precision = $TP / (TP + FP)$ = **87.5%**

Sensitivity = $TP / (TP + FN)$ = **87.5%**

F1 Score: **87.5%**

Specificity = $TN / (TN + FP)$ = **87.5%**

Classification Model

Training Loss and Accuracy :

- 15 total epochs
- Loss in decreasing trend
- Accuracy in Increasing trend
- Accuracy plateaus at 95%
- Loss plateaus at 0.16
- Model working well

Validation Loss and Accuracy:

- 15 epochs and similar trends to training results
- Loss fluctuates around 0.3 mark
- Accuracy remains constant at 87-91% range
- Max accuracy: 94%
- Working well predicting new data

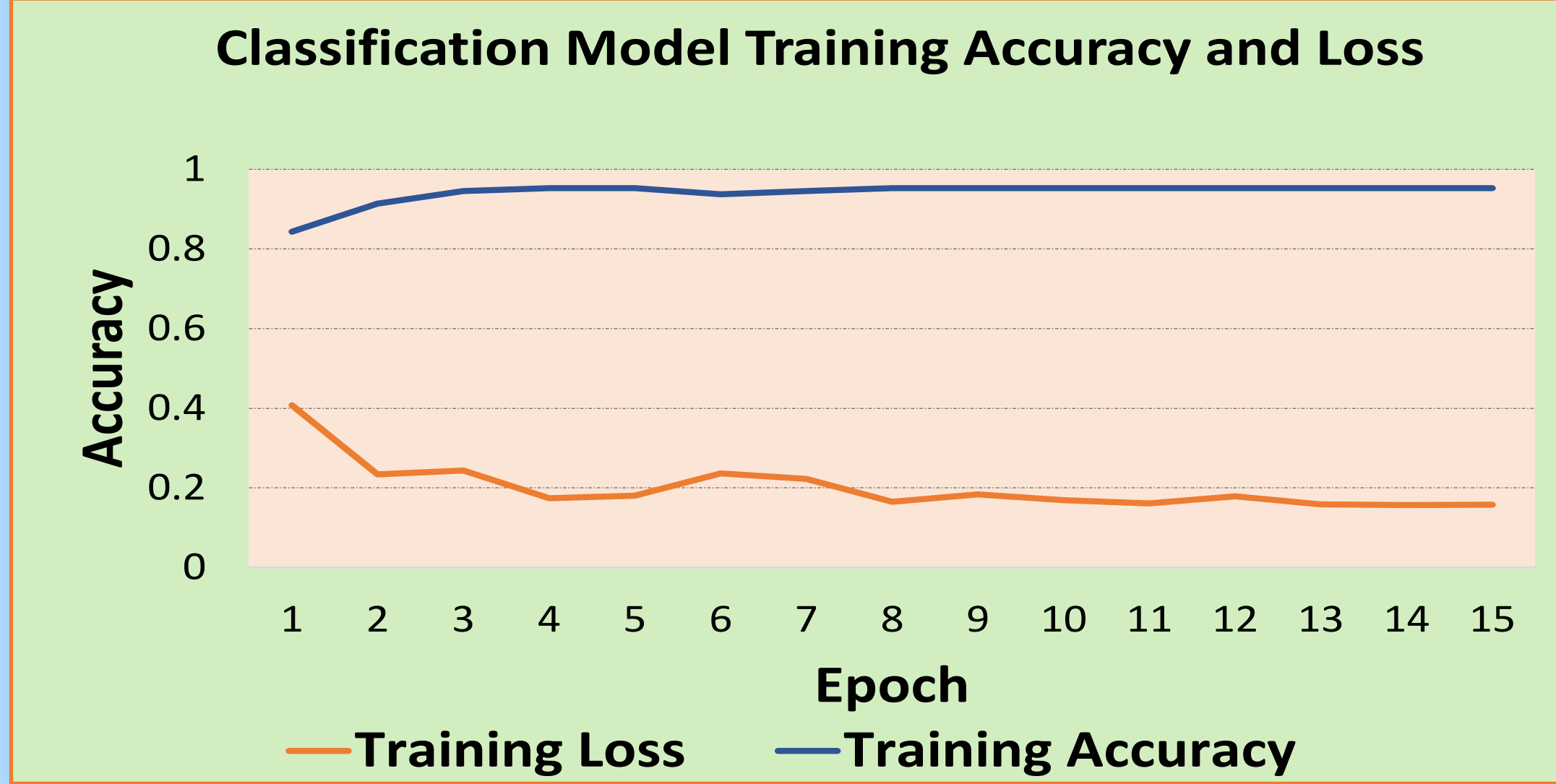


Figure 15. Classification Model Results- Training Loss and Accuracy

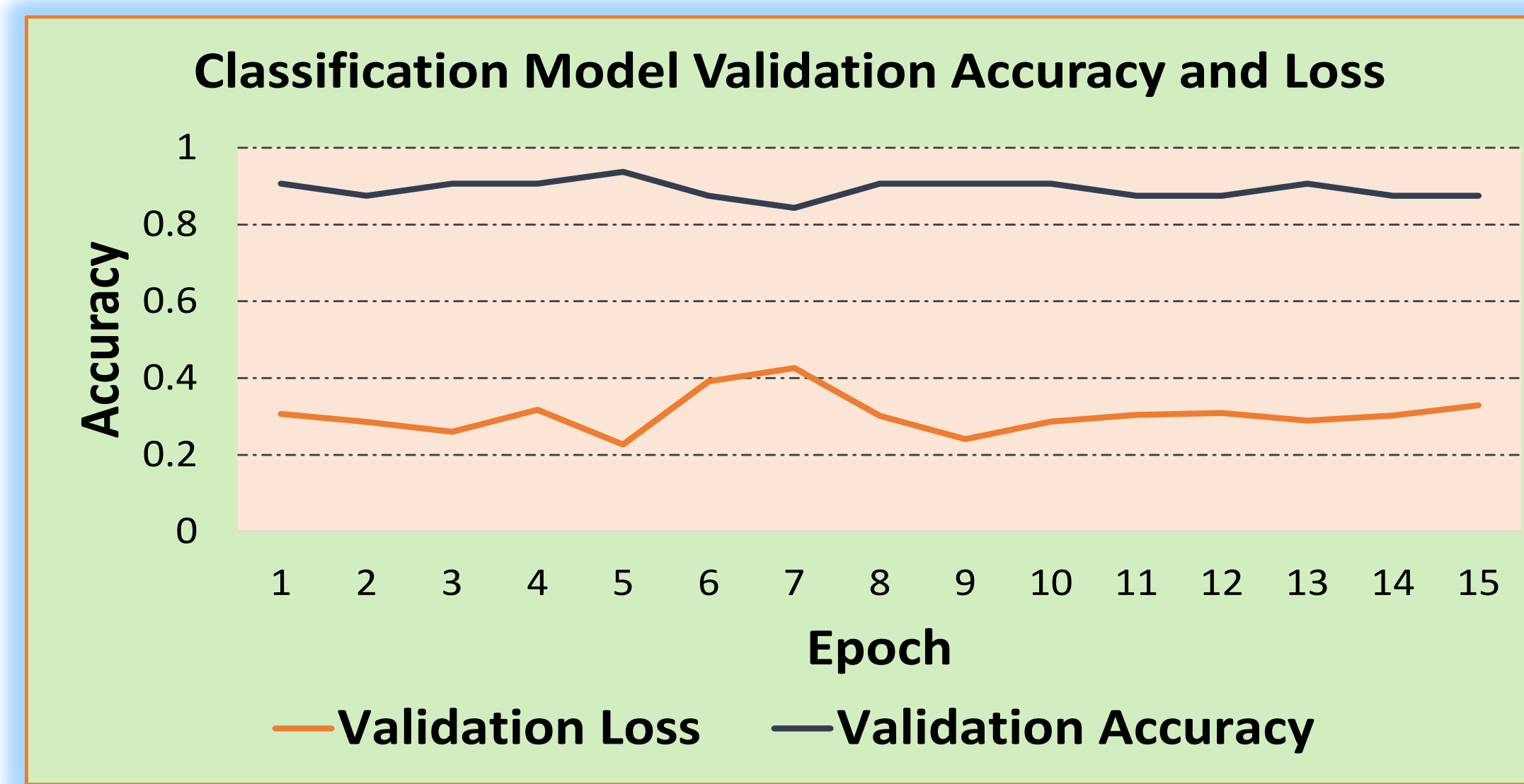


Figure 16. Classification Model Results- Validation Loss and Accuracy

Challenges

- Representative Data collection, model overfitting and model selection.
- While the final accuracy of the classification model was 95%, this number can significantly improve if some aspects of the algorithm are changed. For example, a better dataset might result in more optimal results for the model as the images present in the dataset are easier to segment and classify.
- Another shortcoming could be that the model was overfitting for both the segmentation and classification model. A way to go about this problem is to add dropout layers to the algorithm or to expand the training data.

Conclusion and Path Forward

- The training and validation losses decreased consistently throughout the epochs, indicating that the model learned to classify the images accurately
- The accuracy of the classification model was achieved as high as 95%, which outperforms current early detection tools
- AI integrated workflow has potential to enhance efficiency and reduce errors for radiologists, and to improve patient outcome
- The dataset of images plays a crucial role in determining a model's accuracy. The better-quality pictures may produce results even higher accuracy
- Early detection by deep learning can revolutionize the PDAC early detection, thus can have significant clinical implications

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